

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1488

Tamarisk 230 kV
July 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1488. The Interconnection Customer proposes a 642.1 MW solar photovoltaic facility interconnecting at the Tamarisk 230 kV substation in the Ohio area. The request was received on June 03, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 371 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,027,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tamarisk - Osawatomie 230kV Line	Bluestem Substation - Bodo 230kV Line	126.3	526	93.5
Bluestem - Bodo 230kV	Osawatomie Substation - Tamarisk 230kV	120.5	498	90.0
Bodo Substation - Tamarisk 230kV	Osawatomie - Tamarisk Substation 230kV	106.1	558	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tamarisk - Osawatomie 230kV Line	Rebuild line with higher-capacity conductor	\$10,342,000	100% (\$10,342,000)
Bluestem - Bodo 230kV	Rebuild line with higher-capacity conductor	\$13,487,000	91% (\$12,256,000)
Bodo Substation - Tamarisk 230kV	Reconductor line	\$3,429,000	100% (\$3,429,000)
	Total allocated to project		\$26,027,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.